

// Array [10, 10, 20, 30, 50]

System.out.println(args[0]);

System.out.println(args[1]);

System.out.println(args[2]);

Class: 7

21/04/2025

Run Configuration

used to pass argument!

LOOPS

(Repeated task).

Control + }

Control - }

to change font size.

* 1. FOR Loop

2. For Each Loop

3. While Loop

4. Do While Loop

* works only with booleans.

* For Loop:

Ex:

package com;

public class practice {

public static void main (String[] args) {

System.out.println("Hello world");

System.out.println("Hello world");

System.out.println("Hello world");

System.out.println("Hello world");

System.out.println("Hello world");

}

(8)

outputs: Hello world
Hello world
Hello world
Hello world
Hello world.

* For Loop: Used when you know how many times you have to do task. (If we know Range). It is also known as Entry Controlled Loops.

Syntax:

for (↓
1; 2; 3

It works with 3 things.

1. Initialization (variable) [int a = 0;]

2. Condition (a < 5)

3. update the value.

⇒ for (initialisation; condition; ~~update~~ update the value) { }

ex: for (int a = 1; a <= 3; a++) (8)
(a) ++a

→ a = 1 ⇒ a <= 3 → true → Enter

→ a = 2

→ a = 3

→ a = 4

false exit.

ex: package com;

public class practice {

public static void main (String[] args) {

for (int i = 0; i <= 5; i++) {

System.out.println(i + " Hello world");

}

}

}

int a = 1;

for (; a <= 3; a++)

(8)

int a = 1;

for (; a <= 3;) {

a++;

}

(8)

int a = 1;

for (; ;) {

~~break~~

if (i == 10) {
break;

}

a++;

}

output: 1 Hello world

2 Hello world

3 Hello world

4 Hello world

5 Hello world.

* for (int i = 100; i >= 1; i--) { }
(8) --i

* While Loop: Used when you don't know how many times we have to do task.

int i = 1;

while (i >= 3) {

i++;

Syntax:

while (condition)

* It is also called Entry Controlled Loop }

* Condition - to enter & exit the loop.

Ex: package com;

```
public class practice {  
    public static void main (String[] args) {
```

```
        int i = 1;
```

```
        while (i <= 5) {
```

```
            System.out.println (i + " Hellowld");
```

```
            i++
```

```
        }
```

```
    }
```

```
}
```

outputs: 1. Hellow world

2 Hellow world

3 "

4 "

5 "

Ex: Do while: It executes atleast once irrespective of condition.
It is also called Exit Controlled loop.

Syntax:

```
do {
```

→ [Logic will post in do block only]

```
}
```

```
while (Condition);
```

Ex: int i = 1;

```
do {
```

```
    syso (i);
```

```
    i++;
```

```
while (i <= 3);
```

outputs:

1

2

3

Ex: public class Test {

```
    psvm }
```

```
    Scanner sc = new Scanner (System.in);
```

```
    syso ( "Enter a number?");
```

```
    "Enter how many times you want print");
```

```
    int times = sc.nextInt();
```

```
    for (int i = 1; i <= times; i++) {
```

```
        syso (" i");
```


while loop: upto when??

ex: 2
Scanner SC = new Scanner(System.in);
int times = SC.nextInt();
while (times >= 0) {
 System.out.println("Times");
 times = SC.nextInt();
}

output:

Enter number:
1
1 Times
2
2 Times
3
3 Times
...
0 (program stopped).

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* Nested For loop:

→ one for-loop inside another for-loop.

→ Syntax:

for (initialisation; condition; updation) {

for (initial; cond; upd) {

}

}

Ex:
*
for (int i = 1; i <= 5; i++) {
 print(i);
 for (int j = 1; j <= 5; j++) {
 print(j);
 }
}

output:

i = 1
j = 1
j = 2
j = 3
j = 4
j = 5
i = 2
j = 1
j = 2
j = 3
j = 4
j = 5

→ 1st iteration is done
↳ all iteration will do

Ex: package com;

```
public class NestedForLoop {
```

```
    public static void main(String[] args) {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            sysout("i : " + i);
```

```
            for (int j = 1; j <= 5; j++) {
```

```
                sysout("j : " + j);
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

output:

i : 1

j : 1

j : 2

j : 3

j : 4

j : 5

i : 2

j : 1

j : 2

j : 3

j : 4

j : 5

i : 3

j : 1

j : 2

j : 3

j : 4

j : 5

i : 4

j : 1

j : 2

j : 3

j : 4

j : 5

i : 5

j : 1

j : 2

j : 3

j : 4

j : 5

*

Ex: package com;

```
public class Nater ForLoop {
```

```
    public static void main(String[] args) {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            for (int j = 1; j <= 5; j++) {
```

```
                sysout("*");
```

```
                sysout(""); → (Empty println)
```

```
            }
```

```
        }
```

```
        sysout("\n"); → (Empty println)
```

```
    }
```

```
}
```

Output:

*
Ex:

```
for (int i=1; i<=5; i++) {  
    for (int j=1; j<=i; j++) {  
        syso(" *");  
    }  
    syso(" ");  
}
```

output:

```
 *
 * *
 * * *
 * * * *
 * * * * *
```

Ex:

```
for (int i=5; i>=1; i--) {  
    for (int j=1; j<=i; j++) {  
        syso(" *");  
    }  
    syso("\n");  
}
```

output:

```
 * * * * *
 * * * *
 * * *
 * *
 *
```

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Ex:

```
Scanner sc = new Scanner(System.in);  
String input = sc.nextLine();  
for (int i=0; i<3; i++) {  
    syso(input.charAt(i));  
}
```

output:

```
FLM
F
L
M
```

```
String input = sc.nextLine();  
int length = input.length();  
for (int i=0; i<input.length; i++) {  
    syso(input.charAt(i));  
}
```

output:

```
FRONT
F
R
O
N
T
```


* Break - Continue:

Break: exit from all cases.

→ In loops, if you use break, it will exit from the loop.

```
for (int i=1; i<=100; i++) {
```

```
    if (i==4) {
```

```
        break;
```

```
    }
```

```
    print(i);
```

i=1 - print

i=2 - print

i=3 - print

i=4 - skip (break the loop).

Continue: It skips the current / present iteration.

```
for (int i=1; i<=100; i++) {
```

```
    if (i==4) {
```

```
        continue;
```

```
    }
```

```
    print(i);
```

```
}
```

i=1 → print

i=2 - "

i=3 - "

i=4 - Do not print

i=5 - print

i=6 - "

!

i=100 = print.

Ex:

①

```
package com;
```

```
public class break {
```

```
    psvm {
```

```
        for (int i=1; i<=10; i++) {
```

```
            ← syso(i) if (i==4) {
```

```
                break;
```

```
            }
```

```
        } syso(i);
```

Output:

1
2
3
4

Output: 1
2
3

②

```
for (int i=1; i<=10; i++) {
```

```
    if (i==5) {
```

```
        break continue;
```

```
    }
```

```
    syso(i);
```

```
}
```

Output:

1
2
3
4
5
6
7
8
9
10

→ 5 skips


```

③ for (int i=1; i<=10; i++) {
    if (i==5) {
        syso(i);
        Continue(i==5);
    }
}

```

output:
1
2
3
4
5
6
7
8
9
10

Examples:

```

① package com;
    public class Practice1 {
        Psum {
            for (int i=1; i<=10; i++) {
                if (i>=3 && i<=7) {
                    syso(i); ← Continue;
                }
                syso(i);
            }
        }
    }

```

output:
1
2
8
9
10

output:
3
4
5
6
7

(3, 4, 5, 6, 7 → skips)

```

② for (int i=1; i<=200; i++) {
    if (i%8 == 0) {
        Continue;
    }
    syso(i);
}

```

output:
1
2
3
4
5
6
7 → 8 skips.
9
10
11
12
13
14
15 → 16 skips
17
18
19
20

```

③ for (int i=1; i<=100; i++) {
    if (i%9 == 0) {
        break;
    }
    syso(i);
}

```

output:
1
2
3
4
5
6
7
8

③ package com;
public class Nested Break {

PSVM {

for (int i=1; i<=3; i++) {

for (int j=1; j<=3; j++) {

if (j==2) {

break;

}

sysout("i: " + i + " j: " + j)

}

}

}

}

Output:

i: 1 j: 1 ✓

~~i: 1 j: 2 × (break)~~

i: 2 j: 1 → break.

i: 3 j: 1 → break.

j: 2 → break.

Labels: (used only in loops)

LABELS

① ↳ to break main loop (i.e, outer loop) [we can give any name to this loop].

outer: (label)

for (int i=1; i<=3; i++)

{ for (int j=1; j<=3; j++) {

if (j==2) {

break outer;

}

sysout("i: " + i + " j: " + j);

output:

i=1, j=1 ✓

~~→ i=2, j=2 ×~~

↳ outer loop breaks.

② outer: (label)

for (int i=1; i<=3; i++)

{ inner: for (int j=1; j<=3; j++) {

if (j==2) {

break inner; (label)

}

sysout("i: " + i + " j: " + j);

Output:

i: 1 j: 1

i: 2 j: 2

i: 3 j: 1

Ex:

[Prime number: a number is divisible by 1 & itself. (8)]

package com;

has factors 1 & itself]

public class PrimeNumber {

public static void main (String [] args) {

Scanner sc = new Scanner (System.in);

int num = sc.nextInt(); → syso ("Enter a number :")

int Count = 0;

for (int i = 1; i <= num; i++) {

if (num % i == 0) {

Count++;

}

}

if (Count == 2) {

syso ("prime Numbers")

else {

syso ("Not a Prime Number");

}

}

output: Enter a number:

5

Prime Numbers.

⇒ Enter a number:

4

Not a Prime Number.

Class: 10

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Ex:

package com;

public class Task {

public static void main (String [] args) {

Scanner sc = new Scanner (System.in);

syso ("Enter a number:");

int num = sc.nextInt();

for (int i = 1; i <= 20; i++)

syso (num + " x " + i + " = " + num * i)

}

}

output: Enter a number:

5

5th Table